

6 Lab 5: Lap Data Analysis with WinTAX4

6.1 Introduction

This laboratory activity focuses on the analysis of racing motorcycle acceleration data collected on Phillip Island circuit. The data was acquired via sensors and accelerometers installed on the vehicle and processed using **WinTAX4**, a telemetry software suite developed by Marelli Motorsport widely used in professional motorsport for both real-time and post-processing analysis.

The primary objective of this study is to **compare raw and filtered acceleration data** to determine the most suitable datasets for subsequent analysis. Additionally, the telemetry will be evaluated to **identify the specific corners of the track where the motorcycle experiences the highest lateral acceleration**.

6.2 Exercise 1: Acceleration Data Comparison

6.2.1 Reference System Transformation

The longitudinal *AccLon* and lateral *AccLat* accelerations of the motorcycle are computed by starting from signal acquisition performed by an onboard electronic sensor system utilized to measure and record a vehicle's dynamic behavior: the **IMU (Inertial Measurement Unit)**.

The IMU consists of a redundant **triaxial accelerometer** and a redundant **triaxial gyroscope**.

- The **accelerometers** directly measure the specific forces (linear accelerations) experienced by the vehicle along its structural axes, labeled as *IMU_AccX* (longitudinal), *IMU_AccY* (lateral), and *IMU_AccZ* (vertical).
- The **gyroscopes** measure the angular velocities around these axes, which are subsequently integrated to calculate the precise orientation angles of the motorcycle: **roll** (θ) and **pitch** (ϕ).



Figure 101: Marelli advanced IMU

The raw acceleration data provided by the IMU is inherently bound to the **Vehicle (IMU) Frame** ($X_{IMU}, Y_{IMU}, Z_{IMU}$), meaning the coordinate system moves and rotates continuously alongside the motorcycle.

In high-performance motorcycle dynamics, analyzing raw data directly introduces significant errors due to vehicle articulation (such as leaning during cornering or pitching under heavy braking). For instance, when the motorcycle leans aggressively into a turn, the vehicle's vertical axis (Z_{IMU}) is no longer aligned with gravity. Consequently, the gravity vector projects onto the lateral (Y_{IMU}) and longitudinal (X_{IMU}) sensors, corrupting the true acceleration readings with gravitational components.

To isolate the genuine forces acting on the vehicle relative to the track surface, a kinematic transformation is strictly required. By utilizing the calculated roll (θ) and pitch (ϕ) angles within the **WinTAX4** environment, the acceleration components are mathematically rotated from the vehicle-fixed frame into an **Inertial Reference Frame** ($X_{Inertial}, Y_{Inertial}, Z_{Inertial}$).

In this reference system:

- The $Z_{Inertial}$ axis remains consistently perpendicular to the ground (counteracting gravity).
- The $X_{Inertial}$ and $Y_{Inertial}$ axes lie perfectly parallel to the track surface.

This transformation is essential to extract the true longitudinal acceleration ($AccLon$) and lateral acceleration ($AccLat$), enabling an accurate and uncorrupted analysis of the motorcycle's handling performance, tire grip utilization, and lap dynamics.

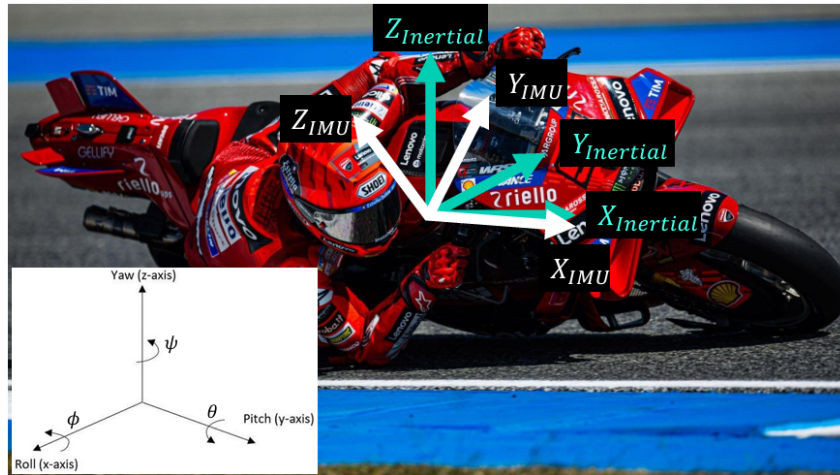


Figure 102: Vehicle and inertial frames

The mathematical procedure to transform the acceleration components measured in the IMU vehicle-fixed frame into the inertial ground-fixed frame is based on coordinate rotation matrices. According to the vehicle's telemetry configuration, the orientation is defined by two primary angles:

- **Pitch (θ):** Rotation around the vehicle's lateral axis (Y -axis).
- **Roll (ϕ):** Rotation around the vehicle's longitudinal axis (X -axis).

The transformation follows a strict sequential order: first, a pitch rotation (θ around the Y -axis), followed by a roll rotation (ϕ around the X -axis). These rotations are defined within the fixed reference frame (vehicle reference frame).

The general transformation equation relating the inertial acceleration vector a_{Inertial} to the measured vehicle acceleration vector a_{Vehicle} is given by:

$$a_{\text{Inertial}} = R \cdot a_{\text{Vehicle}} \quad (38)$$

Where the respective acceleration vectors are defined as:

$$a_{\text{Inertial}} = \begin{bmatrix} Acc_{\text{Lon}} \\ Acc_{\text{Lat}} \\ Acc_{\text{Ver}} \end{bmatrix}, \quad a_{\text{Vehicle}} = \begin{bmatrix} AccX_{\text{IMU}} \\ AccY_{\text{IMU}} \\ AccZ_{\text{IMU}} \end{bmatrix} \quad (39)$$

The total rotation matrix R is obtained from the product of the individual elementary rotation matrices, respecting the specified sequence:

$$R = R_{\text{Roll}}(\phi) \cdot R_{\text{Pitch}}(\theta) \quad (40)$$

The elementary rotation matrices for pitch (R_{Pitch}) and roll (R_{Roll}) are structured as follows:

$$R_{\text{Pitch}}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \quad (41)$$

$$R_{\text{Roll}}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \quad (42)$$

Substituting the elementary matrices into the transformation equation yields the full matrix mapping:

$$\begin{bmatrix} Acc_{\text{Lon}} \\ Acc_{\text{Lat}} \\ Acc_{\text{Ver}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \cdot \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \cdot \begin{bmatrix} AccX_{\text{IMU}} \\ AccY_{\text{IMU}} \\ AccZ_{\text{IMU}} \end{bmatrix} \quad (43)$$

Performing the matrix multiplication gives the complete rotation matrix operator:

$$\begin{bmatrix} Acc_{\text{Lon}} \\ Acc_{\text{Lat}} \\ Acc_{\text{Ver}} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ \sin(\phi) \sin(\theta) & \cos(\phi) & -\sin(\phi) \cos(\theta) \\ -\cos(\phi) \sin(\theta) & \sin(\phi) & \cos(\phi) \cos(\theta) \end{bmatrix} \cdot \begin{bmatrix} AccX_{\text{IMU}} \\ AccY_{\text{IMU}} \\ AccZ_{\text{IMU}} \end{bmatrix} \quad (44)$$

By expanding the system of algebraic equations from the matrix multiplication, we obtain the explicit expressions for each inertial acceleration component, which can be implemented within the **WinTAX4 VCH (Virtual CHannels) math functions**:

$$Acc_{\text{Lon}} = \cos(\theta) AccX_{\text{IMU}} + \sin(\theta) AccZ_{\text{IMU}} \quad (45)$$

$$Acc_{\text{Lat}} = \sin(\phi) (\sin(\theta) AccX_{\text{IMU}} - \cos(\theta) AccZ_{\text{IMU}}) + \cos(\phi) AccY_{\text{IMU}} \quad (46)$$

$$Acc_{\text{Ver}} = -\cos(\phi) \sin(\theta) AccX_{\text{IMU}} + \sin(\phi) AccY_{\text{IMU}} + \cos(\phi) \cos(\theta) AccZ_{\text{IMU}} \quad (47)$$

6.2.2 Comparison performed on WinTAX

Using the equations showed in the previous section, it is possible to compute the accelerations in the inertial frame through the math VCHs. The comparison is carried out considering:

- the **raw acceleration data**.
- the **acceleration data filtered with a cut-off frequency $f_c = 10$ Hz**.

The filtering process is performed via a **low-pass filter**, which removes high-frequency noise caused by engine vibrations and track roughness. This smooths the signal, allowing for an accurate analysis of the motorcycle's true trajectory and peak accelerations without sensor distortions. Filtering on WinTAX4 is realized with the command `IIR_LP`.

The following pictures shows the examples of the VCHs of raw and filtered longitudinal acceleration. The same approach was implemented for both lateral and vertical accelerations with the respective equations.

The screenshot shows the WinTAX4 VCH editor interface. The 'Name' field is 'AccLon_raw', 'Frequency' is 'Auto', 'Format' is 'Auto', 'Decimals' is 'Auto', and 'Unit' is 'g'. The 'Expression' field contains the formula: `IMU1_AccX*cos(IMU_Pitch*pi/180) + IMU1_AccZ*sin(IMU_Pitch*pi/180)`. The 'Comment' field is empty. The 'Show' checkbox is checked. Below the expression field, there is a small icon of a function $f(x)$.

Figure 103: Raw longitudinal acceleration in inertial frame

The screenshot shows the WinTAX4 VCH editor interface. The 'Name' field is 'AccX_Filt', 'Frequency' is 'Auto', 'Format' is 'Auto', 'Decimals' is 'Auto', and 'Unit' is 'g'. The 'Expression' field contains the formula: `IIR_LP(IMU1_AccX,10)`. The 'Comment' field is empty. The 'Show' checkbox is checked. Below the expression field, there is a small icon of a function $f(x)$.

Figure 104: Filtered longitudinal acceleration in vehicle frame

The screenshot shows the WinTAX4 VCH editor interface. The 'Name' field is 'AccLon_Filt', 'Frequency' is 'Auto', 'Format' is 'Auto', 'Decimals' is 'Auto', and 'Unit' is 'g'. The 'Expression' field contains the formula: `AccX_Filt*cos(IMU_Pitch*pi/180) + AccZ_Filt*sin(IMU_Pitch*pi/180)`. The 'Comment' field is empty. The 'Show' checkbox is checked. Below the expression field, there is a small icon of a function $f(x)$.

Figure 105: Filtered longitudinal acceleration in inertial frame

6.2.3 Final Plots and Conclusions

The obtained results were plotted on graphs comparing raw and filtered accelerations in function of time analyzed on a single lap.



Figure 106: Longitudinal and lateral acceleration plots

Based on the generated telemetry plots, the **filtered signal** is significantly more appropriate for vehicle dynamics analysis than the raw signal.

A direct comparison between the raw data (green trace) and the data processed with a 10 Hz low-pass filter (red trace) highlights the following phenomena:

- **Overestimation of Acceleration Limits:** In the longitudinal acceleration plot (*AccLon*), the raw sensor data registers a maximum positive peak of 2.01351 g. However, the filtered signal reveals that the actual physical acceleration peak experienced by the vehicle is only 1.47363 g. The discrepancy of over 0.54 g is entirely composed of high-frequency noise.
- **Corruption of Maximum Lateral Forces:** A similar effect is observable in the lateral acceleration plot (*AccLat*). The raw channel captures an artificial maximum cornering peak of 2.30977 g, whereas the filtered trace accurately isolates the true physical limit of the vehicle at 1.89594 g.
- **Engineering Justification:** The high-frequency noise visible in the raw traces originates from non-kinematic sources such as engine vibrations and track surface roughness (e.g., curbs and asphalt imperfections). Relying on raw data would lead to a severe overestimation of tire grip utilization and vehicle performance. The 10 Hz low-pass filter successfully removes these parasitic frequencies, making the filtered channels (*AccLon_Filt* and *AccLat_Filt*) the only reliable choice for statistical lap reporting and engineering analysis.

To provide an indisputable mathematical justification for selecting the filtered signal over the raw sensor data, a **Fast Fourier Transform (FFT) analysis** was conducted to evaluate the Power Spectral Density (PSD) of both channels.

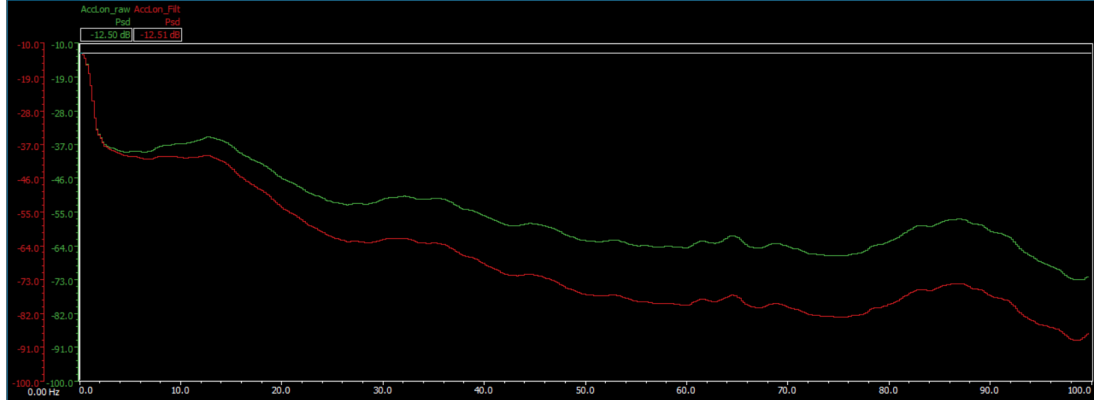


Figure 107: PSDs of raw and filtered longitudinal acceleration

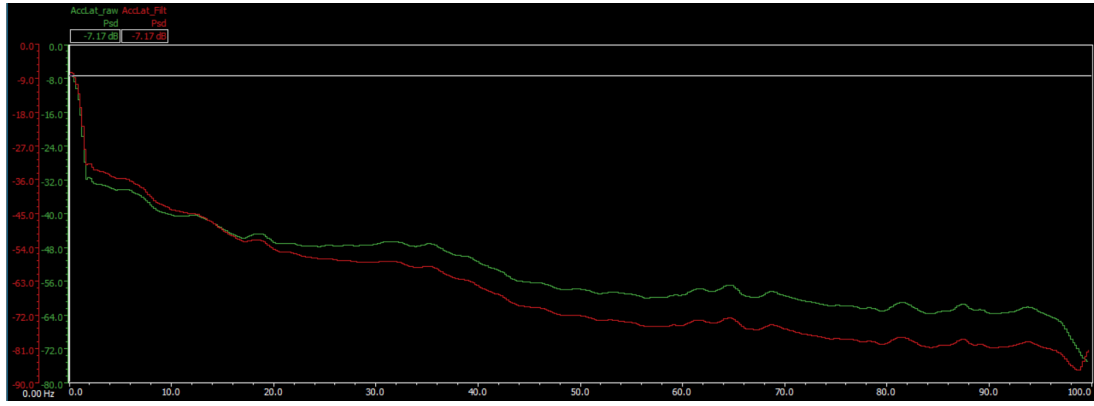


Figure 108: PSDs raw and filtered lateral acceleration

The frequency spectrum, ranging from 0 Hz to 100 Hz, illustrates the precise mathematical impact of the applied 10 Hz low-pass filter:

- **Low-Frequency Domain (0 Hz to 10 Hz):** In this region, the raw signals (green traces) and the filtered signals (red traces) overlap almost perfectly for both longitudinal and lateral components. This demonstrates that the filter preserves the integrity of the signal's energy associated with the true macro-dynamics, handling maneuvers, and cornering trajectories of the motorcycle, which naturally occur at low frequencies.
- **High-Frequency Attenuation (10 Hz to 100 Hz):** Beyond the 10 Hz cutoff threshold, the red filtered curves drop significantly below the green raw curves. This consistent attenuation gap across the high-frequency band proves that the filter successfully dampens parasitic energy and high-frequency noise originating from engine vibrations and track surface roughness.

To verify whether the filtering process was executed correctly, a **correlation plot** between the raw and filtered acceleration signals was analyzed.

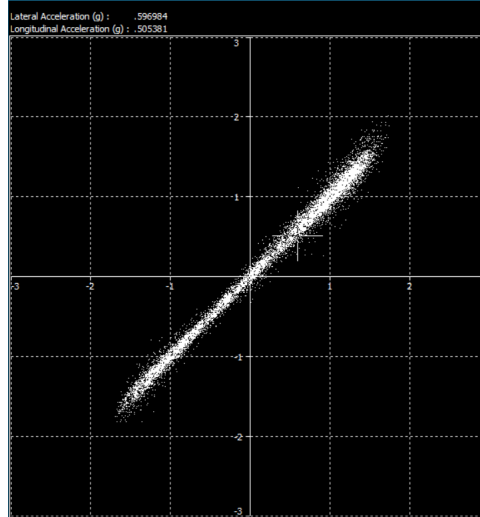


Figure 109: Correlation plot between raw and filtered lateral acceleration

This scatter plot demonstrates an optimal correlation between the two channels; the extremely narrow and well-defined diagonal band confirms that high-frequency noise has been effectively suppressed while preserving a clean, high-fidelity signal. Furthermore, the lack of any elliptical separation proves the complete absence of detrimental phase lag or time delay, validating the overall quality and responsiveness of the filter.

Regarding the vehicle's dynamic limits and traction envelope, the **corrected G-G plot** of filtered lateral versus longitudinal acceleration was evaluated.

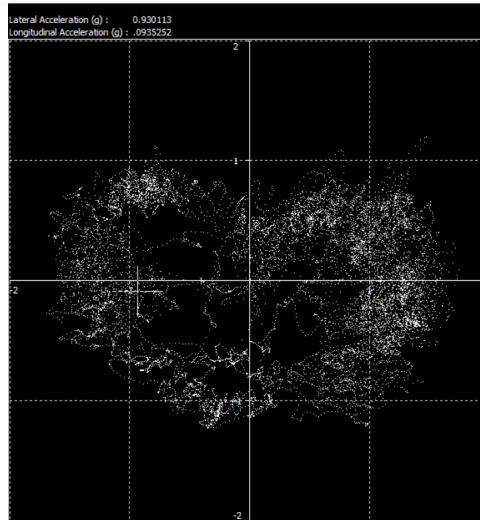


Figure 110: G-G plot between filtered accelerations

The diagram shows a well-defined, heart-shaped envelope characteristic of motorcycle dynamics, where longitudinal acceleration is physically constrained during maximum cornering. Thanks to effective filtering and proper sensor calibration, the boundaries of the friction ellipse are distinct and free of high-frequency noise dispersion. Furthermore, the data cloud is correctly centered around the origin, validating the sensor's zeroing and proving that the filtered signals accurately replicate the real-time forces acting on the chassis.

6.3 Exercise 2: Lap Report for Maximum Lateral Acceleration

The objective of this exercise is to analyze filtered telemetry data to determine the point of maximum lateral acceleration on circuit. By evaluating the comprehensive lap report, the specific corner where the vehicle experiences the peak lateral inertial force can be identified.

6.3.1 Telemetry Evaluation and Circuit Layout

The spatial context of the analysis is established through the track configuration. The following picture shows the layout of the Phillip Island circuit, including the numbering of the turns used as references throughout the telemetry logs.



Figure 111: Phillip Island circuit

The primary analysis was conducted by referencing the WinTAX4 lap report. The following picture shows the summary visualization from the software, displaying the overall lateral acceleration trends alongside the lap summary metrics and maximum value tables.

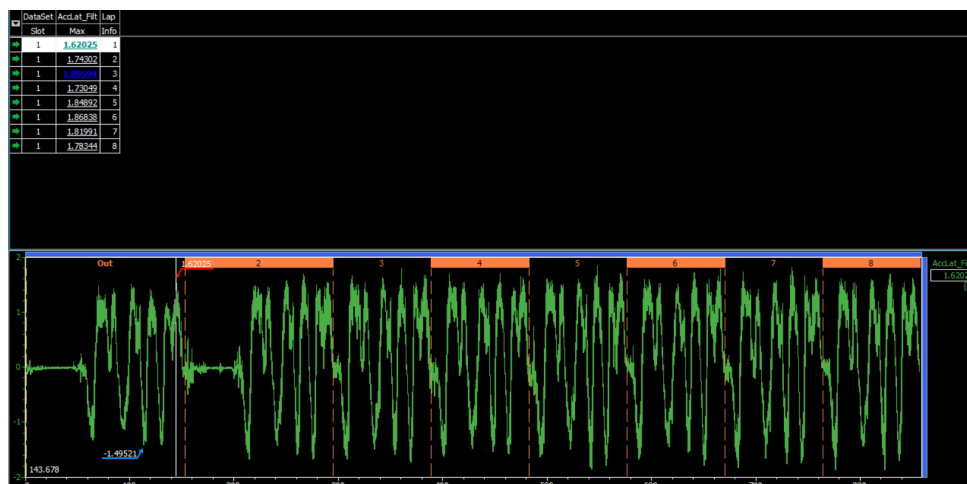


Figure 112: Lap report for maximum lateral acceleration

6.3.2 Results and Discussion

By extracting the critical parameters from the filtered telemetry data across the monitored stints, the specific performance metrics per lap were compiled. Table 10 summarizes the analyzed laps, detailing the maximum filtered lateral acceleration recorded and isolating the exact corner where the absolute peak occurred.

Lap	Max Lateral Acceleration (g)	Turn
1	1.62025	Turn 12
2	1.74302	Turn 9
3	1.89594	Turn 9 Exit
4	1.73049	Turn 9 Exit
5	1.84892	Turn 9 Exit
6	1.86838	Turn 12 Exit
7	1.81991	Turn 9 Entry
8	1.78344	Turn 9

Table 23: Maximum lateral acceleration per lap and corresponding turn Location

The peak of lateral acceleration consistently alternates between the high-speed sections of Turn 9 and Turn 12. Turn 9 typically dominates due to the significant vertical load variations on its uphill compression, where minor changes in driving lines and tire thermal states shift the filtered peak between entry, mid-corner, and exit. Conversely, the shifts to the high-velocity arc of Turn 12 (as seen in Laps 1 and 6) can be justified by unoptimized initial tire temperatures or localized momentum losses at Turn 9, which naturally transfer the absolute lap's peak of lateral acceleration to the final high-speed corner.

NOTE: Unfortunately, we faced problems due to bugs of the track window. The cursor icon of the driver did not move along the circuit, so we were not able to determine the right point of maximum lateral acceleration on the track in this way. So we tried an alternative solution in which the point of interest about the acceleration was identified comparing the lateral acceleration channel `AccLatFilt` with the traveled track distance channel `sLapVehicle`. The value of distance found corresponds to a specific point on the circuit according to its length.

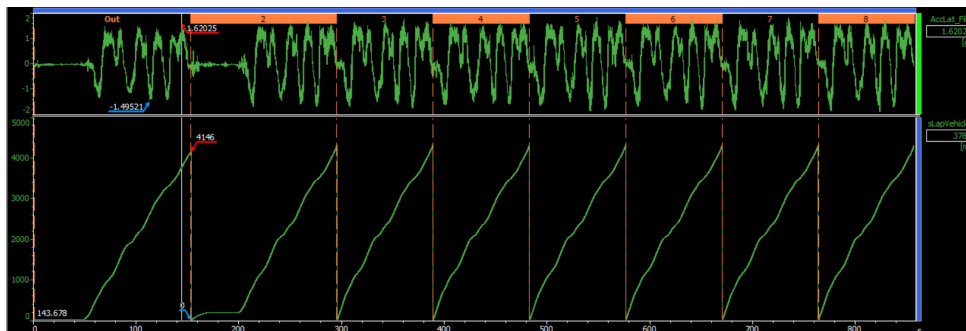


Figure 113: Lateral acceleration and traveled distance comparison

6.4 Mugello Circuit

The same telemetry analysis procedure conducted for Phillip Island has been replicated for the Mugello circuit. The objective of this exercise is to analyze the filtered telemetry

data to determine the point of maximum lateral acceleration on the track. By evaluating the comprehensive lap report, the specific corner where the motorcycle experiences the peak lateral inertial force can be identified.

6.5 Telemetry Evaluation and Circuit Layout

The spatial context of the analysis is established through the track configuration. Figure 114 shows the layout of the Mugello circuit, including the numbering of the turns used as references throughout the telemetry logs.

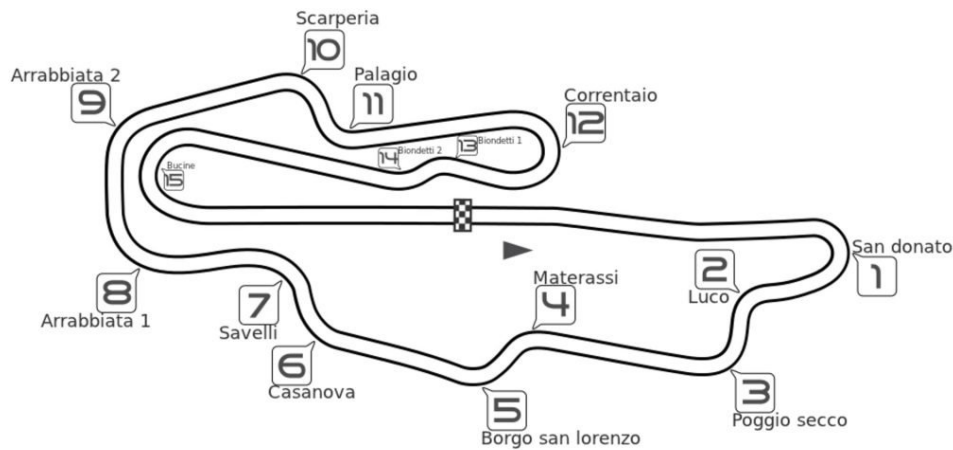


Figure 114Mugello circuit layout

Before analyzing the lap report, the filtering process was validated. As performed previously, a Fast Fourier Transform (FFT) analysis was conducted to evaluate the Power Spectral Density (PSD) of the acceleration channels. The application of a 10 Hz low-pass filter successfully dampens the high-frequency noise originating from engine vibrations and track surface roughness.

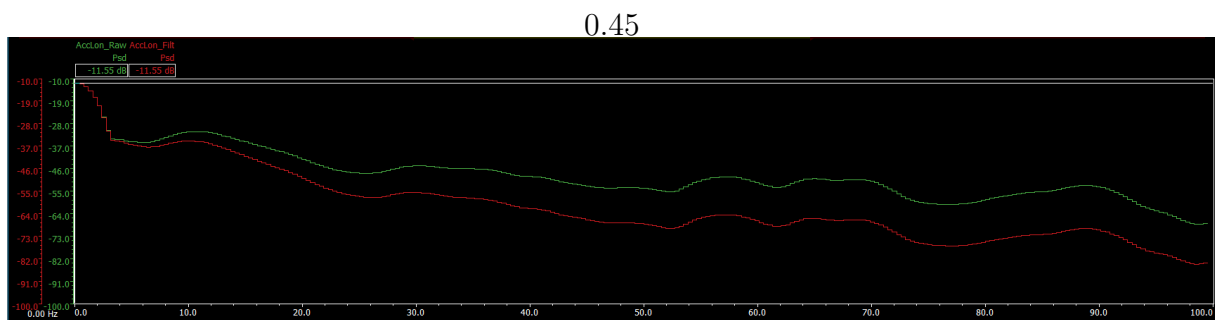


Figure 115PSD of raw and filtered longitudinal acceleration

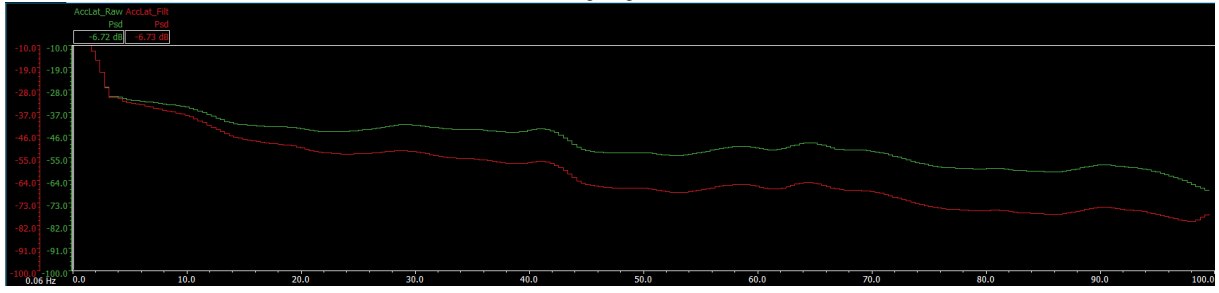


Figure 116 PSD of raw and filtered lateral acceleration

To verify whether the filtering process was executed correctly, a correlation plot between the raw and filtered lateral acceleration signals was analyzed. The resulting scatter plot demonstrates an optimal correlation between the two channels; the narrow and well-defined diagonal band confirms that high-frequency noise has been effectively suppressed while preserving a clean signal. Furthermore, the lack of any elliptical separation proves the absence of detrimental phase lag or time delay.

Regarding the vehicle's dynamic limits and traction envelope, the corrected G-G plot of filtered lateral versus longitudinal acceleration was also evaluated. The diagram shows the well-defined envelope characteristic of motorcycle dynamics, where longitudinal acceleration is physically constrained during maximum cornering. Thanks to effective filtering, the boundaries of the friction ellipse are distinct and the data cloud is correctly centered.

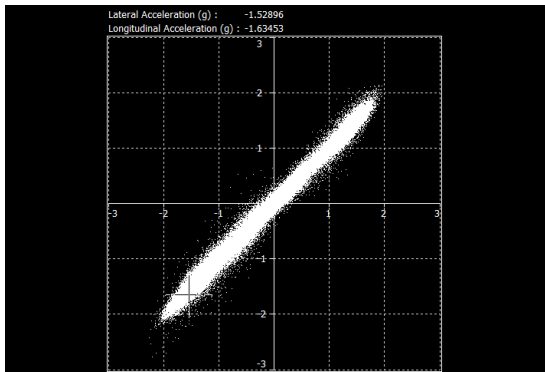


Figure 117 Correlation plot between raw and filtered lateral acceleration

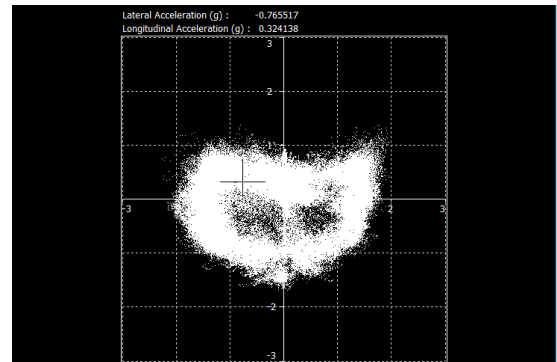


Figure 118 G-G plot between filtered accelerations

6.5.1 Results and Discussion

The primary analysis was conducted by referencing the WinTAX4 lap report. Figure 119 shows the summary visualization from the software, displaying the overall lateral acceleration trends alongside the lap summary metrics.



Figure 119Lap report for maximum lateral acceleration (Mugello)

By extracting the critical parameters from the filtered telemetry data across the monitored stints, the specific performance metrics per lap were compiled. Table 24 summarizes the analyzed laps, detailing the maximum filtered lateral acceleration recorded and isolating the exact corner where the absolute peak occurred.

Lap	Max Lateral Acceleration (g)	Turn
1	1.92132	Exit turn 15
2	1.9522	Exit turn 15
3	1.89765	Exit turn 15
4	1.95201	Turn 13
5	1.89964	Turn 11
6	1.92424	Exit turn 15
7	1.7684	Turn 7
8	1.8273	Turn 7

Table 24Maximum lateral acceleration per lap and corresponding turn location (Mugello)